An Improved Ellipse Propagation Model for Location Estimation in facilitating Ubiquitous Computing

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Abstract

Mobile location estimation is a crucial technology for ubiquitous computing. A directional propagation model - the Ellipse Propagation Model (EPM) is proposed by our research group for locating a mobile station (MS) within a radio cellular network with an accuracy that can enable a number of location based services to realize ubiquitous computing. EPM assumes the contour line of signal strength resembles an ellipse with the base station situates at one of the focus. By using a Geometric Algorithm, the location of the mobile station can be estimated. However, since one parameter in our Geometric Algorithm is fixed, errors may be induced as the surrounding environment changes. In view of this, we would like to propose a new algorithm - the Iterative Algorithm to estimate the location of mobile station based on EPM. With the technical support of two local mobile phone operators, we have conducted a series of experiments using real data and experiment results showed that the proposed Iterative Algorithm outperforms the Geometric Algorithm by a good margin of 18% in terms of average error.

1 Introduction

Recently, mobile location estimation is receiving considerable attention in the field of wireless communications. Many positioning technologies have been developed. The most famous location system is The Global Positioning System (GPS) is one of the location systems that is mature enough and commercially available [1, 2]. Other proposed location estimation methods include Time-Of-Arrival (TOA), Time Difference Of Arrival (TDOA), Enhanced Observed Time Difference (E-OTD) and Angle-Of-Arrival (AOA) [3]. These positioning technologies are based on timing information or angular information. Time based methods, such as, TOA, TDOA and E-OTD, calculate the distance between the Mobile Station (MS) and the Base Station (BS) by measuring the propagation time of the signal and multiply it by the speed of light. By using trilateration, the position of the MS can be estimated. On the other hand, angular approaches, like AOA, measure the angle between the MS and the BS and then estimated the location of the MS by using triangulation. Although these positioning technologies are simple, these approaches are only applicable to CDMA system since it can provide the timing or angular information. However, quite a number of countries have adopted the GSM network instead of the CDMA network. And the GSM network can only provide the loss of signal strength due to signal attenuation [4].

Since the loss of signal strength is the common attribute of all radio cellular network, thus location estimation algorithms proposed here are applicable to all radio cellular network for ubiquitous computing.

Our group has also proposed several location estimation approaches based on the received signal strength (RSS) [5, 6, 7]. However, these methods have not included the directional properties of the antenna. From our observations, we found that the BSs have directional properties. That is, BSs always transmit signal in a direction. In view of that, our research group has proposed an Ellipse Propagation Model (EPM) in [8]. EPM is derived from the original propagation model [3]. We observed that the antenna transmits the signal in a direction. Thus, the contour line of signal strength should not be a circle. We therefore modify the original propagation model by considering the contour line of signal strength as an ellipse with the BS sitting at one of the focuses. Since the RSS is the only attribute we have, therefore EPM focuses on the relationship between the MS-BS distance and the RSS. We also proposed a Geometric Algorithm to estimate the location of the MS based on EPM. Experiment results

have proven that EPM using the Geometric Algorithm is very encouraging. However, one of the parameters in the Geometric Algorithm is fixed. This may cause some defects in the accuracy since the parameters may vary in different environment. Moreover, the Geometric Algorithm does not have self-modification property, namely, the estimation of the Geometric Algorithm is unstable. In view of this, we would like to propose a new approach to estimate the location of the MS with EPM - the Iterative Algorithm. The Iterative Algorithm has self-modification property. It chooses a convergence value as the estimation. And our experiment results have proven that the Iterative Algorithm is superior to the Geometric Algorithm with a trade off between accuracy and computational cost.

This paper is divided into five sections. In the following section, we will depict some location estimation algorithms proposed by our group. In section 3, the Iterative Algorithm will be presented. Afterwards, we will discuss the simulation results of the Iterative Algorithm using real data. And at last, the conclusion and future work for our research will be presented.

2 Related Works

Previously, our group has proposed two algorithms for location estimation, namely, the Center of Gravity (CG) algorithm and the Circular Trilateraion (CT) algorithm [5, 7]. Both CG and CT are making use of the RSS for estimating the position of the MS. They assumed the relationship between the MS-BS distance (d) and the RSS (s) is $s \propto d^{-2}$ based on the inverse square law [9]. However, due to the interference and distortion by buildings, the relationship is remodelled into $s \propto d^{-\alpha}$, where α is a variable being related to the environment.

We have also proposed an Ellipse Propagation Model (EPM) and a Geometric Algorithm to provide the location of the MS in [8]. The Ellipse Propagation Model (EPM) considers the antenna directional transmission property and assumes the contour line of the signal strength for one antenna as an ellipse which the base station is on one of the focuses. While the Geometric Algorithm is a simple and useful method to provide the location of the MS; it is derived from the CT algorithm and to reduce the defects of the CT algorithm.

2.1 Center of Gravity (CG)

The CG approach defines the location estimation formula as,

$$\begin{cases} x = \frac{x_1 s_1^{-\alpha} + x_2 s_2^{-\alpha} + x_3 s_3^{-\alpha} + \ldots + x_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \\ y = \frac{y_1 s_1^{-\alpha} + y_2 s_2^{-\alpha} + y_3 s_3^{-\alpha} + \ldots + y_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \end{cases} \tag{2.1}$$

where (x, y) is the estimated location of the MS. $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ are the locations of n receiving BSs. $s_1, s_2, ..., s_n$ are the corresponding RSS from each BS [5].

Although the CG approach has proven its outstanding performance in metropolitan area, it can only estimate a mobile device inside the convex hull of the BSs involved.

2.2 Circular Trilateration (CT)

The basic idea of the CT approach is to construct 3 circles with the RSS for 3 different BSs. By knowing the location of the three BSs and the mapping between RSS and MS-BS distance, the intersection of these 3 circles is the estimated location of the MS. Similar to CG, CT models the relationship between the MS-BS distance (d) and the RSS (s) as $d \propto (N + s)^{-\alpha}$, where N is the normalization constant. By making use of this relationship, we can constructed 3 circles as follows,

$$\begin{cases} (x - x_1)^2 + (y - y_1)^2 = (\frac{k}{s_1{}^{\alpha}})^2 \\ (x - x_2)^2 + (y - y_2)^2 = (\frac{k}{s_2{}^{\alpha}})^2 \\ (x - x_3)^2 + (y - y_3)^2 = (\frac{k}{s_3{}^{\alpha}})^2 \end{cases} \tag{2.2}$$

where, s_1, s_2 and s_3 are the RSSs from 3 receiving BSs with their geographic locations as (x_1, y_1), (x_2, y_2) and (x_3, y_3) respectively and k be a common scaling factor. The location of the MS is then estimated as the intersection point of these 3 circles [7].

Although CT does not have the convex hull problem, it does not always provide an estimation. This is because intersection may not always appear due to signal fading. Moreover, the CG and the CT approaches do not take the transmission direction of the BS into account, which is not realistic. Thus, we have designed a new approach, the EPM which is an improvement of the CT algorithm.

2.3 The Ellipse Propagation Model

From our observations, we found that the BSs have directional transmission property. The antenna transmits the signal in some directions. That is, the antenna transmits the largest power in one direction,

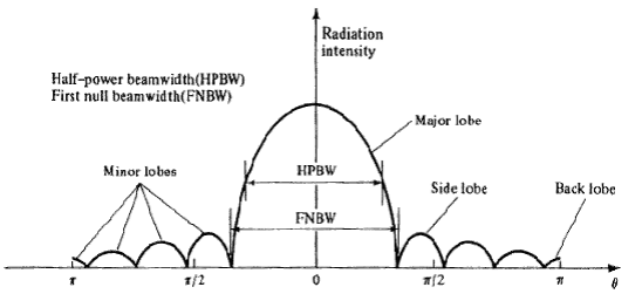

Figure 1. The directional antenna

while transmits small or none power in other directions. We can plot the contour line of signal strength around an antenna as **Figure 1**. As shown in the Figure 1, the contour line of the signal strength for the directional antenna is not a circle, which violates the assumption of the original propagation model in free space.

We assume the contour line of the signal strength as an ellipse instead of a circle based on our studies. The EPM adopts a different relationship between the MS-BS distance and the RSS. We can use the following mathematical formula to depict the EPM.

$$d = k(s_0/s)^{1/\alpha}(1-e)/(1-e\cos(\theta)) \tag{2.3}$$

where

d is the distance between MS and BS;
k is the proportion constant;
s_0 is the transmitting power of the BS;
s is the signal power received;
e is the eccentricity value of the ellipse, it describes the figure of the signal strength contour line;
θ is the deviation between the ellipse principal axis and the line of MS and BS;
α is called the path loss exponent.

We call this relationship as the Ellipse Propagation Model (EPM), where the contour line of the signal strength is an ellipse [8]. The Ellipse Propagation Model (EPM) can be illustrated in **Figure 2**.

The EPM has four parameters: k, α , e and θ. We consider the parameters of k and α as the region parameters, and the parameter e is used to describe the figure of the ellipse, while the deviation θ is parameter for each MS. We can use the field test data to find out the values of these parameters, then translate the signal strength into two points distance. That is the main idea of the EPM.

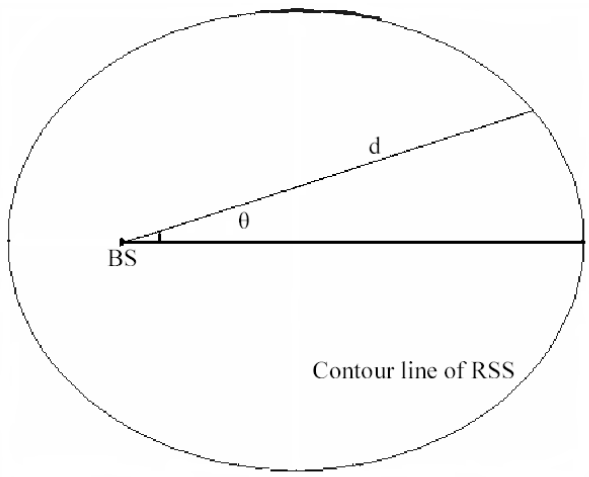

Figure 2. The Ellipse Propagation Model(EPM)

2.4 The Geometric Algorithm under EPM

The Geometric Algorithm is used to provide the estimation of the location of the MS. We choose the value of deviation between the ellipse principal axis and the line of the BS and the center (or the weighted center) of the BSs locations as the estimation of θ. So the EPM exactly has three parameters: k, α and e.

Suppose a MS receives RSS, s_1, s_2, s_3 from three BSs with locations, (x_1, y_1), (x_2, y_2), (x_3, y_3) respectively. In addition, the distances between the MS and the BSs are denoted by $d(s_1)$, $d(s_2)$, $d(s_3)$, sometimes, and they are simply denoted by d_1,d_2 and d_3. Thus, by the formula of the two points distance in the 2-D Euclidian space, we can form three circles formulas which are shown as follows,

$$\begin{cases} (x-x_1)^2 + (y-y_1)^2 = d_1^2 \\ (x-x_2)^2 + (y-y_2)^2 = d_2^2 \\ (x-x_3)^2 + (y-y_3)^2 = d_3^2 \end{cases} \tag{2.4}$$

Basically, the geometric interpretation of this equation group means three circles in 2-D space, and the solution is the intersection point of these circles. However, if an intersection point does not exist, we may not be able to provide an estimation for the MS. In order to solve this problem, the Geometric Algorithm can not solve this equation group directly. Instead, the Geometric Algorithm derives another three equation group from the equation group (2.4). As each new equation group can provide a solution, thus, we will have three solutions. The estimation of the MS will then be the center of these three solutions [8].

The estimation of the Geometric Algorithm is:

$$\begin{aligned} x &= (2m(y_3 - y_1) - 2n(y_2 - y_1))/|A| \\ y &= (-2m(x_3 - x_1) + 2n(x_2 - x_1))/|A| \end{aligned} \tag{2.5}$$

where

$|A| = 4[(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)]$;
$m = [d_1^2 - (x_1^2 + y_1^2)] - [d_2^2 - (x_2^2 + y_2^2)]$;
$n = [d_1^2 - (x_1^2 + y_1^2)] - [d_3^2 - (x_3^2 + y_3^2)]$;
(x_1, y_1),(x_2, y_2) ,(x_3, y_3) are the BSs locations.

3 The Iterative Algorithm under EPM

In our previous research, we have proposed a EPM and a Geometric Algorithm to provide the location of the MS [8]. We based on these results to further develop an Iterative Algorithm under EPM. The basic idea of the Iterative Algorithm is to improve the Geometric Algorithm to provide a more accurate and more stable estimation.

The Geometric Algorithm based on EPM can always provide an estimation whenever the MS receives three or more antenna signals. However, the deviation between the major transmitting direction and the line of MS-BS, θ, in the Geometric Algorithm is fixed and is dependent on the location of the MS. Hence, we present a self-modification method illumined by the Iterative Algorithm to provide the location of the MS and the value of deviation θ. We name this method as the Iterative Algorithm. Since the θ in Geometric Algorithm is an approximate value only, some error may be induced in the estimation. Thus, the Iterative Algorithm should have better accuracy than the Geometric Algorithm by eliminating these errors.

The EPM with the Iterative Algorithm has four parameters: k, α, e and θ. k and α are the region parameters, and e is the parameter to describe the shape of the contour lines. They all can be provided by training with the field test data. Thus, parameter θ and the location of the MS can be calculated with the Iterative Algorithm. In the Iterative Algorithm, we do not use a constant value to estimate θ, but update the estimation of θ for each iteration. Since the value of θ depends on the location of the MS, the estimation of the location of the MS can be used to calculate the value of θ.

We choose an initial value to calculate the value of θ. Then, the distance between MS and BS is derived, and a solution is calculated by the Geometric Algorithm. This solution is a new location of the MS. For each iterative computing, we update the values of θ and the location of the MS. If the series of the locations of the MS have a convergence value, we choose the stabilized convergence value as the estimation of the location of the MS.

3.1 Structure of the Iterative Algorithm

Let (x, y) be the coordinates of the location of the MS, and x, y are two independent random variables. We consider the estimation of the location of the MS as a conditional expectation of the RSS and the locations of BSs, denoted by,

$$(x', y')^T = E((x, y)^T | x_0, y_0; s; l) \tag{3.1}$$

where x_0 and y_0 are the location of the MS we want to find; s is the information of RSS, and l is the location information of BSs; x' and y' are the estimation of the location of the MS, and they are random variables too.

By rewriting the exact location of the MS with x and y as two parts, we have,

$$\begin{cases} x = f(x_0, y_0; s; l) + \epsilon \\ y = g(x_0, y_0; s; l) + \eta \end{cases} \tag{3.2}$$

where $f(x_0, y_0; s, l)$ and $g(x_0, y_0; s, l)$ are the certain terms; x_0 and y_0 are the estimations of x and y; ϵ and η are random variables; f and g are some functions structures, which have first order and second order derivatives. Furthermore, we assume that $E(\epsilon) = 0$, $E(\eta) = 0$, $var(\epsilon) = \sigma_\epsilon^2$, $var(\eta) = \sigma_\eta^2$, $cov(\epsilon, \eta) = 0$, x_0 and y_0 are unbiased estimations of x and y. Thus,$E(x - x_0) = 0$,$E(y - y_0) = 0$.

We choose the certain terms as the estimation of the location of the MS.

$$\begin{cases} x_0 = f(x_0, y_0; s; l) \\ y_0 = g(x_0, y_0; s; l) \end{cases} \tag{3.3}$$

Since x_0 and y_0 appear in both sides, the iterative method can be considered to provide the solution. We call the equation group (3.3) as the structure of the Iterative Algorithm.

Therefore, by picking an initial value, for example, the estimation from the CG algorithm (x_{CG}, y_{CG}), we can provide the estimation of the MS if the iterative formulas are convergent.

3.2 Using Geometric Algorithm with the Iterative Algorithm

Suppose the MS, with location (x, y), received RSS, s_1, s_2, s_3 from three BSs at locations, $l_1(\alpha_1, \beta_1)$, $l_2(\alpha_2, \beta_2)$, $l_3(\alpha_3, \beta_3)$ and the output powers, o_1, o_2, o_3 respectively. In addition, the distances between the MS and the BSs are denoted by d_1, d_2, d_3.

We depict the Iterative Algorithm by using the Geometric Algorithm structure based on EPM. We can get the following formulas in [8]

$$\begin{cases} x = (2m(\beta_3 - \beta_1) - 2n(\beta_2 - \beta_1))/|A| \\ y = (-2m(\alpha_3 - \alpha_1) + 2n(\alpha_2 - \alpha_1))/|A| \end{cases} \tag{3.4}$$

Then we define the deviation θ_l as

$$\theta_l = \begin{cases} \frac{5\pi}{2} - bear_l - \arccos(\frac{x-\alpha_l}{\sqrt{(x-\alpha_l)^2+(y-\beta_l)^2}}) & \text{if } y > \beta_l \\ \frac{\pi}{2} - bear_l + \arccos(\frac{x-\alpha_l}{\sqrt{(x-\alpha_l)^2+(y-\beta_l)^2}}) & \text{if } y \le \beta_l \end{cases}$$

where θ_l is the deviation which contains the bearing information; $bear_l$ is the bearing information; (x, y) is the MS location; (α_l, β_l) is the BS location.

So the iterative formulas become,

$$\begin{aligned} x_{n+1} &= 2[(\beta_3-\beta_2)(d_1^2(n)-(\alpha_1^2+\beta_1^2)) \\ &\quad +(\beta_1-\beta_3)((d_2^2(n))-(\alpha_2^2+\beta_2^2)) \\ &\quad +\beta_2-\beta_1)(d_3^2(n)-(\alpha_3^2+\beta_3^2))]/|A| \\ y_{n+1} &= 2[(\alpha_2-\alpha_3)(d_1^2(n)-(\alpha_1^2+\beta_1^2)) \\ &\quad +(\alpha_3-\alpha_1)((d_2^2(n))-(\alpha_2^2+\beta_2^2)) \\ &\quad +(\alpha_1-\alpha_2)(d_3^2(n)-(\alpha_3^2+\beta_3^2))]/|A| \end{aligned}$$

Thus,

$$\begin{cases} x_{n+1} = f(x_n, y_n; s_1, s_2, s_3; l_1, l_2, l_3) \\ y_{n+1} = g(x_n, y_n; s_1, s_2, s_3; l_1, l_2, l_3) \end{cases} \tag{3.5}$$

Eq.(3.5) is called the iterative formulas. Given an initial value, we can derive a series of (x_n, y_n) from the iterative formulas. If (x_n, y_n) converges to one point $(\hat{x}, \hat{y})$, then $(\hat{x}, \hat{y})$ is considered to be the location estimation of the MS.

3.3 Convergence of the Iterative algorithm

We define two distance functions for our discussion: one is a distance based on EPM, the other is a common distance in a 2-D Euclidean space, denoted by,

$d(x, y, \alpha_l, \beta_l, p_{epm}, s_0, s) = \frac{k(s_0/s)^{1/\alpha}(1-e)}{1-e\cdot\cos(\theta(x,y,\alpha_l,\beta_l,bear_l))}$

$de(x, y, \alpha_l, \beta_l) = \sqrt{(x-\alpha_l)^2+(y-\beta_l)^2}$

where $d(x, y, \alpha_l, \beta_l, p_{epm}, s_0, s)$ and $de(x, y, \alpha_l, \beta_l)$ are two distance functions, and $p_{epm} = (k, \alpha, e, \theta)$.

To simplify our discussion, we fix one variable when the other changes, for example, we fix y for x, then we denote them as $d(x)$ and $de(x)$. We assume the estimated distance between the MS and the ith BS using EPM is $d_i(x, y)$. If these three circles intersect at one point, then $d(x, y) = de(x, y)$. Otherwise, we assume $d(x, y) \le de(x, y)$.

Theorem 3.1 *If (α_1, β_1), (α_2, β_2), (α_3, β_3) are not in the same line, and suppose that $|\frac{d_i^2(x,y)}{de_i^2(x,y)} \cdot \frac{(y-\beta_i)}{de_i(x,y)} \cdot \frac{e_i\sin(\theta_i)}{1-e_i\cos(\theta_i)}| \le 1$, $|\alpha_i - \alpha_j| \ge 4$, if $|\alpha_i - \alpha_j| \ne 0$; $|\beta_i - \beta_j| \ge 4$, if $|\beta_i - \beta_j| \ne 0$; where $i \ne j$ and $i, j = 1, 2, 3$. Then $\{x_n\}$ and $\{y_n\}$ converge.*

Proof: For our convenience discussion. We assume that $\alpha_1 \le \alpha_3 \le \alpha_2$ and $\beta_2 \le \beta_1 \le \beta_3$. If $\alpha_1 = \alpha_3$, we add the condition, $\frac{\beta_1-\beta_2}{\beta_3-\beta_2} \le \frac{1}{2}$.

Set

$$A = \begin{pmatrix} 2(\alpha_2-\alpha_1), & 2(\beta_2-\beta_1) \\ 2(\alpha_3-\alpha_1), & 2(\beta_3-\beta_1) \end{pmatrix}$$

$$b = \begin{pmatrix} (d_1^2-(\alpha_1^2+\beta_1^2))-(d_2^2-(\alpha_2^2+\beta_2^2)) \\ (d_1^2-(\alpha_1^2+\beta_1^2))-(d_3^2-(\alpha_3^2+\beta_3^2)) \end{pmatrix}$$

We rewrite the equation as the matrix formula, $AX = b$, where $X = (x, y)^T$.

Since (α_1, β_1),(α_2, β_2) ,(α_3, β_3) are not in the same straight line, so $det(A) \ne 0$. And the solution is $X = A^{-1}b$. Then we rewrite it as an iterative formula, $X_{n+1} = A^{-1}b(X_n)$.

Based on the definition of EPM,

$k(o_i/s_i)^{1/\alpha}\frac{1-e_i}{1+e_i} \le d_i(n) \le k(o_i/s_i)^{1/\alpha}$, $i = 1, 2, 3$.

So both $\{x_n\}$ and $\{y_n\}$ have **bounds**.

We fix y when x_n changes, and fix x when y_n changes for our convenience discussion. In this paper, the convergence of $\{x_n\}$ will be discussed in detail while $\{y_n\}$ is a similar case.

Define

$f(x) = (\beta_3-\beta_2)d_1^2(x)+(\beta_1-\beta_3)d_2^2(x)+(\beta_2-\beta_1)d_3^2(x)$ and $f(x)$ is a continuous function, then we obtain:

$(x_{n+1} - x_n)det(A)/2 = f(x_n) - f(x_{n-1})$.

By **the Lagrange Theory** [10], $f(x)$ has first derivative, we have: $f(x_n) - f(x_{n-1}) = f'(x)(x_n - x_{n-1})$, where x is between x_n and x_{n-1}.

$$\begin{aligned} \frac{f'(x)}{2} &= [\frac{(\beta_3-\beta_2)\cdot d_1^2(x)\cdot(y-\beta_1)\cdot e_1\sin(\theta_1)}{de_1^3(x)\cdot(1-e_1\cos(\theta_1))} \\ &\quad +\frac{(\beta_1-\beta_3)\cdot d_2^2(x)\cdot(y-\beta_2)\cdot e_2\sin(\theta_2)}{de_2^3(x)\cdot(1-e_2\cos(\theta_2))} \\ &\quad +\frac{(\beta_2-\beta_1)\cdot d_3^2(x)\cdot(y-\beta_3)\cdot e_3\sin(\theta_3)}{de_3^3(x)\cdot(1-e_3\cos(\theta_3))}] \end{aligned}$$

Set $K = 2f'(x)/det(A)$, and

$det(A) = (\alpha_2-\alpha_1)(\beta_3-\beta_1)-(\alpha_3-\alpha_1)(\beta_2-\beta_1)$.

If $\alpha_3 - \alpha_1 \ne 0$ then $\alpha_3 - \alpha_1 \ge 4$.

By the theory condition,

$|K| < \frac{2}{(\alpha_2-\alpha_1)} + \frac{2}{\alpha_3-\alpha_1} \le \frac{4}{\alpha_3-\alpha_1} \le 1$

If $\alpha_3 - \alpha_1 = 0$,

$|K| < \frac{2}{(\alpha_2-\alpha_1)-(\alpha_2-\alpha_1)\frac{\beta_1-\beta_2}{\beta_3-\beta_2}} \le \frac{4}{\alpha_2-\alpha_1} \le 1$

That is , $|K| < 1$, and

$(x_{n+1} - x_n) = K(x_n - x_{n-1})$,

So $\{x_n\}$ converges. Similarly, $\{y_n\}$ converges too. Q.E.D.

4 Simulation Results

With the technical support of two mobile operators in Hong Kong, we have conducted an intensive field test in many regions in Hong Kong in order to validate

our model. We have divided the data into two parts: 30% of the data for training the models, and 70% of the data for estimating the location of the MS. We choose the distance between the exact location and the estimation location as the error criteria to describe the estimation accuracy.

With divided our experiment in two phases: the first phase is to train the model using 30% of the field test data, and the second phase is to estimate the location of the MS with the rest of the 70% of the field test data. With the effect of signal attenuation, our method sometimes provides an unreasonable solution. Hence, we use an anchor point to select the estimation of the location of the MS. If our estimation is not within a radius of 1000 meter with respect to the anchor point, we discard our solution and consider the our method fail to provide the estimation of the location of the MS. We choose the exact location of the MS as the anchor point in our model training phase. And we choose a weighted center of the locations of the BSs which we received signal from as the anchor point for estimating the location of the MS. So the missing ratio of our method includes two cases: One case is the number of BSs we received signal from is less than three, which does not meet our model assumption; the other case is the solution provided by our method is not within the predefined area in radius.

The data we collected from the field test are first used to provide the EPM parameters. These parameters are then put into a ***Lookup Table*** which will be used during the testing process. In the testing phase, we apply the Geometric Algorithm and the Iterative Algorithm to compute the field test data, then calculate their errors for the MS estimation. Lastly, we compared the results of the Iterative Algorithm with the results of CG, CT and Geometric Algorithm.

4.1 Estimating the EPM parameters

We choose 30% of the field test data to find out the parameters of EPM. For saving the computation cost, we divide all types of BSs into three groups for each region, which are denoted by Macro, Micro and others. Since the environment condition is similar within the same region, we assume the value of path loss exponent, α, and the proportion value k are the same throughout a region. Thus, there will be five parameters need to be trained in each region, e_1, e_2, e_3, k and α, where e_1, e_2 and e_3 are the EPM parameters for Macro, Micro and others respectively within one region. We set the step of the eccentricity, e_1, e_2 and e_3 as 0.1, and the step of path loss exponent, α, is 0.1 , while the step of the proportion value k is 0.05 for saving the computational cost. As α is expected to be within a range as suggested by [3], therefore, we choose α to vary within a range of 3 and 10 in order to meet the situation of Hong Kong and the proportion value is between 0.5 and 1.5. While the eccentricity of an ellipse has its natural limitation, it can vary within a range of 0 and 1 only. We provide the values of these parameters for each region. The results are shown in the ***Lookup Table***: **Table 1**.

4.2 Results of the Iterative Algorithm

After obtaining the ***Lookup Table***, Geometric Algorithm and Iterative Algorithm are used to estimate the location of the MS.

The Iterative Algorithm is used to estimate the location of the MS. Results are shown in **Table 2**. From **Table 2**, the estimation results using the Iterative Algorithm have good performance in many regions, such as Central, CheungShaWan, KwunTong, Mongkok, PE-MK, PrinceEdward, ShamShuiPo and SheungWan, where the averages of the estimating errors are below 125 meters. And in some regions, the 67% point values are below 125 meters, such as Mongkok and ShamShuiPo. In general, the improvement of the Iterative Algorithm over the Geometric Algorithm is inspiring.

There exist some regions where both the Geometric Algorithm and the Iterative Algorithm are not performing well, such as HungHum and LaiKing. We believe that these regions have special terrains and network layouts which may not fit the EPM directly. For example, in HungHum, we found that the MS always received signals from other regions. Moreover, some areas, like LaiKing, have very serious signal fading problem on the grounds that these regions are in front of the hills. In addition, we noticed that the RRS in these regions did not follow the rule of path loss well. Therefore, Some modifications may be required to readjust the EPM in order to handle these special regions.

4.3 Compare among the CG, CT and Geometric Algorithm

In this experiment, we compare our results with the results of the CG , CT and Geometric Algorithm. We present the mean, the standard deviation of the estimating errors and the success ratio of computing to describe the quality of the estimations. And the results are shown in **Table 3**.

The CG algorithm has the best success ratio for providing the estimation of the location of the MS, but its estimation has the worst performance within these algorithms. Since the CG estimation is just a weighted

Region	e_1	e_2	e_3	k	α	Region	e_1	e_2	e_3	k	α
Aberdeen	0.4	0	0	1.5	7.3	ShamShuiPo	0.4	0.6	0	0.55	8.4
CauseWayBay	0.4	0.9	0	1.45	4.5	ShaTin	0.7	0.7	0	1.45	6.1
Central	0	0.2	0	1.45	7.2	ShekKipMeiPark	0.6	0	0	1.5	6.6
CheungShaWan	0.1	0	0	1.25	8.4	SheungShui	0	0	0	1.5	8.1
FoTan	0.9	0	0	1.5	6.4	SheungWan	0.4	0.9	0	1.45	8.4
HappyValley	0.6	0	0	1.5	8.1	TaiKooShing	0.9	0	0	0.55	4.9
HungHom	0	0	0.1	1.45	7.5	TaiWai	0.3	0.4	0	0.55	5.6
KowloonBay	0.4	0.2	0.9	0.55	6.4	TaiWoHau	0.2	0	0	0.65	5.3
KowloonCity	0.6	0.3	0.5	1.5	6.7	TinShuiWai	0.7	0	0	0.55	5.6
KowloonTong	0	0	0	1.5	7.8	TsingYi	0.1	0.1	0	1.5	7.7
KwaiFong	0.9	0	0	0.55	6.1	TsuenWan	0.9	0.5	0	0.55	8.4
KwunTong	0.9	0	0	1.4	7.5	TszWanShan	0.4	0	0	1.45	7.6
LaiChiKok	0	0.9	0	0.55	6.9	TuenMun	0.3	0	0	1.5	7.3
LaiKing	0.7	0.9	0.7	0.55	5.7	WanChai	0.2	0	0	1.5	7.4
MaOnShan	0.8	0	0	0.55	8.4	WongTaiSin	0.9	0	0	1.5	7.7
Mongkok	0.6	0	0	1.5	6.7	YauTong	0	0	0	1.5	7.1
PE-MK	0.6	0.9	0	1.5	7.8	YauYatChuen	0.3	0	0	1.5	7.5
PrinceEdward	0.9	0.9	0.3	0.55	8.4	YuenLong	0.9	0	0.9	1.5	5.9

Table 1. The Lookup Table

Region	Ave.	Imp.	Std.	67%	90%	Region	Ave.	Imp.	Std.	67%	90%
Aberdeen	228.85	5.26%	135.40	226.30	483.61	ShamShuiPo	103.36	6.79%	55.17	119.61	193.02
CauseWayBay	258.62	28.42%	189.56	323.19	619.47	ShaTin	351.58	5.72%	172.12	401.61	577.97
Central	109.64	4.96%	64.37	143.96	201.92	ShekKipMeiPark	314.36	3.80%	170.07	379.47	570.76
CheungShaWan	120.47	12.45%	66.34	158.94	211.89	SheungShui	507.99	-9.37%	333.80	798.05	1007.03
FoTan	280.15	8.16%	134.60	356.50	460.99	SheungWan	116.10	7.25%	58.98	144.56	190.77
HappyValley	343.89	9.86%	222.31	430.83	714.78	TaiKooShing	250.84	36.26%	153.91	295.97	440.13
HungHom	934.48	-0.27%	528.76	1414.11	1838.55	TaiWai	226.03	3.20%	90.86	269.04	319.49
KowloonBay	199.82	14.3%	108.01	227.17	343.94	TaiWoHau	277.40	-6.97%	144.96	356.22	508.62
KowloonCity	223.78	10.06%	129.99	264.94	550.48	TinShuiWai	384.60	-0.29%	197.60	513.49	779.81
KowloonTong	255.83	13.45%	153.84	209.41	499.45	TsingYi	554.74	-13.17%	262.30	674.04	931.13
KwaiFong	182.95	3.21%	94.83	224.52	318.17	TsuenWan	143.42	8.78%	60.28	138.80	236.72
KwunTong	115.05	8.78%	58.84	126.32	202.43	TszWanShan	232.29	11.86%	135.04	274.57	448.45
LaiChiKok	325.22	2.34%	175.28	214.62	617.10	TuenMun	319.73	5.05%	153.89	421.09	516.47
LaiKing	682.62	14.58%	249.34	996.45	1181.29	WanChai	166.92	16.64%	112.18	182.60	334.99
MaOnShan	353.11	7.27%	116.73	410.16	500.64	WongTaiSin	306.57	-20.35%	147.16	377.40	515.72
Mongkok	88.36	3.49%	50.04	111.08	150.08	YauTong	387.53	5.18%	321.40	484.65	1030.33
PE-MK	110.08	4.68%	62.25	134.99	202.64	YauYatChuen	227.77	1.8%	113.69	278.87	389.61
PrinceEdward	118.29	14.51%	69.91	138.67	214.91	YuenLong	267.59	18.29%	285.83	229.26	976.63

Table 2. Result with Iterative Algorithm and improvement over the Geometric Algorithm (Unit: meter)

Model	**Average Error**	**Improvement %**	**Std.**	**sample number**	**success ratio %**
CG	495.76	27.08%	787.05	116354	96.93%
CT	470.99	23.24%	986.06	116354	76.28%
Geometric Algorithm	441.09	18.04%	721.12	116354	84.08 %
Iterative Algorithm	361.52	0%	541.87	116354	79.13%

Table 3. Compare among the CG, CT, Geometric and Iterative Algorithm and improvement(Unit: meter)

mean of the locations of the BSs which we received the signal from, its recommendation is always within the convex hull formed by these BSs locations, regardless whether the exact location of the MS is inside or outside the convex hull!

The CT algorithm has been proposed to solve the convex hull problem by trilateration. The CT estimation has better performance than the CG estimation does, but the CT algorithm discards some snapshots information for the cost. The CT algorithm has the least success ratio within these algorithms. Since the CT algorithm uses three base stations information to provide the location of the MS, it can not always provide a solution.

The Geometric Algorithm has derived from the CT algorithm. The Geometric Algorithm yet again improves the CT algorithm, since it can always provide a solution for the estimation of the location to the MS.

The results of the Geometric Algorithm has better performance than the CT and CG algorithms based on the average and the standard deviation of the estimation of errors. The success ratio of the Geometric Algorithm is better than that of the CT algorithm.

The Iterative Algorithm has improved the Geometric Algorithm. Since the Geometric Algorithm fixes a parameter of EPM to provide the location of the MS, it reduces the accuracy of estimation of the Geometric Algorithm. Since the Iterative Algorithm needs more extra conditions to guarantee to provide a convergence solution for the estimation, the success ratio of the Iterative Algorithm is some what lower than that of the Geometric Algorithm. But the Iterative Algorithm has the best performance in terms of the average and standard deviation of the estimating errors among these algorithms. Hence, the Iterative Algorithm is the best among these algorithms. However, the Iterative Algorithm needs more computational cost to provide a better estimation of the MS, and this is the trade between computation cost and accuracy. As a whole, the Iterative Algorithm provides an 18% ($\frac{441.09-361.52}{441.09} \times 100\%$) of improvement in terms of average error over the Geometric Algorithm on locating the MS within a radio cellular network.

5 Conclusions and Future work

In this paper, we present an iterative method to estimate the location of a MS under EPM. The Iterative Algorithm is an improvement over the Geometric Algorithm. From our simulation results, we have shown that the Iterative Algorithm has better accuracy than the Geometric Algorithm. Furthermore, we have proven that the Iterative Algorithm is superior to the existing algorithms. However, the Iterative Algorithm requires more computational cost for the implementation.

During this research, we found that signals do fluctuate at the same place. Signal attenuation can be affected by conditions, such as weather and car movement. The fluctuating signals will induce more errors in our estimation. As for our future work, we follow two threads to go forward our research. We will try to find out a filtering method to reduce the effect of signal fluctuation. Moreover, we will extend EPM to provide a more accuracy estimation. EPM is just a simple relationship between the RSS and the MS-BS distance. But it is a crude relationship for the RSS and the distance between MS and BS. And we use the same path loss exponent, α, and proportion value k in one region for saving computational cost, which may increase more error of the estimation. On the other hand, since EPM is a 2-D model only, we should extend it into a 3-D model to meet the real situation for locating the MS and to facilitate location based services. So to provide a filter method to reduce the effect of the signal fluctuation and to extend our EPM into a 3-D model for providing a 3-D location estimation are our future research work.

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